

BGP (Border Gateway Protocol) & Multicast Routing – Lecture Notes

Border Gateway Protocol (BGP)

- Definition: BGP is an inter-domain (EGP) routing protocol used to exchange routing information between Autonomous Systems (AS) on the Internet.
- Type: Path Vector Protocol.
- Purpose: Makes routing decisions based on path, network policies, and rule sets.

Types of Autonomous Systems (AS)

- Stub AS: Connected to only one other AS. Example: A small company connected to a single ISP.
- Multihomed AS: Connected to more than one AS, but does not allow transit traffic. Example: Enterprise network connected to 2 ISPs for redundancy.
- Transit AS: Provides transit service for other ASes. Example: Tier-1 ISP.

CIDR (Classless Inter-Domain Routing)

- Removes dependency on traditional class A, B, C networks.
- Uses prefix notation (e.g., 192.168.1.0/24).
- Reduces size of routing tables by route aggregation.

Types of Attributes in BGP

- Well-Known Mandatory: AS-Path, Next-Hop, Origin.
- Well-Known Discretionary: Local Preference, Atomic Aggregate.
- Optional Transitive: Communities, Aggregator.
- Optional Non-Transitive: MED (Multi-Exit Discriminator).

BGP Session

- BGP routers (peers) establish TCP sessions on port 179.
- Peers exchange routing information.

Types of BGP Session

- External BGP (eBGP) → Between routers in different ASes.
- Internal BGP (iBGP) → Between routers in the same AS.

BGP Packet Format

- BGP messages exchanged over TCP include:
 1. Open Message
 2. Update Message
 3. Keepalive Message

- 4. Notification Message

Open Message

- First message sent to establish a session.
- Contains Version, AS Number, Hold Time, BGP Identifier, Optional Parameters.

Format of Open Message

- +-----+
- | Version | My AS | Hold Time | BGP Identifier |
- +-----+
- | Optional Parameters Length | Parameters |
- +-----+

Update Message

- Used to advertise or withdraw routes.
- Contains Withdrawn Routes, Path Attributes, Network Layer Reachability Information (NLRI).

Keepalive Message

- Ensures session is alive.
- Sent periodically.
- Contains no routing information.

Notification Message

- Sent when an error is detected.
- Terminates the BGP session.
- Example errors: Invalid message, authentication failure.

How BGP Solves Count-to-Infinity Problem

- BGP uses AS-Path Attribute.
- Each AS adds its AS number to the path.
- If a loop occurs, the same AS number will appear again → router detects loop → discards route.

Unicasting

- Communication between one sender and one receiver.
- Example: Sending an email to one user.

Multicasting

- Communication between one sender and multiple specific receivers.
- Example: Online video lecture to registered students.

Broadcasting

- Communication between one sender and all nodes in the network.
- Example: ARP requests in LAN.

Intradomain Routing Protocol

- Used within an AS.
- Examples: OSPF, IS-IS, MOSPF.

MOSPF (Multicast OSPF)

- Extension of OSPF to support multicast routing.
- Uses IGMP (Internet Group Management Protocol) for group membership.
- Adds multicast information to OSPF LSAs.

Multicast Distance Vector (DVMRP)

- Distance Vector Multicast Routing Protocol.
- Builds multicast distribution trees using Reverse Path Forwarding (RPF).
- Example: Early multicast protocol used in MBone.

Core Based Tree (CBT) Protocol

- Builds a shared multicast tree rooted at a core (rendezvous point).
- Reduces overhead compared to source-based trees.

PIM (Protocol Independent Multicast)

- Works with any unicast routing protocol.
- Two modes:
 1. Dense Mode (PIM-DM) → Assumes many receivers (flood and prune).
 2. Sparse Mode (PIM-SM) → Assumes few receivers (uses Rendezvous Point).

MBone (Multicast Backbone)

- Early experimental backbone for multicast on the Internet.
- Tunnels multicast traffic through unicast networks.
- Used in early live video/audio streaming tests.